10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Review Exercise 12 — Probability

Unit 12 | Review + MCQs | New Book (PECTAA)

Review Exercise 12 (Solved)

The review joins all three parts of the unit: the sample space, the addition law and the multiplication law, with and without replacement.

1. Four possible answers are given for the following questions. Choose the correct answer.

(i) The probability of impossible event is:

(a) 0 (b) 1

(c) 2 (d) −1

Answer: (a) 0 — an impossible event has no favourable outcome, so n(E) = 0

(ii) What is the probability of getting a head when a fair coin is tossed once?

(a) 0 (b) 0.25

(c) 0.5 (d) 1

Answer: (c) 0.5 — one head out of two equally likely outcomes = 1/2

(iii) The sum of probabilities of all possible outcomes of an experiment is:

(a) 0.5 (b) 0.25

(c) 1 (d) 0.4

Answer: (c) 1 — the total probability of the whole sample space is 1

(iv) If P(A) = 0.6, then the probability of event A not happening is:

(a) 0.4 (b) 0.6

(c) 1 (d) 1.6

Answer: (a) 0.4 — $P(\bar{A}) = 1 - P(A) = 1 - 0.6 = 0.4$

(v) Two events are said to be mutually exclusive if:

(a) they can happen at the same time. (b) one affects the other.

(c) they cannot happen together. (d) they are always equal.

Answer: (c) they cannot happen together. — mutually exclusive means $P(A \cap B) = 0$

(vi) If one coin is tossed and one dice is rolled, then the number of sample point are:

(a) 3 (b) 6

(c) 2 (d) 12

Answer: (d) 12 — $n(S) = 2 \times 6 = 12$

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(vii) What is the probability of sure event?

(a) 1 (b) 0
(c) 2/3 (d) 4/5

Answer: (a) 1 — a sure event contains the whole sample space

(viii) The probability of getting a number greater than 6 on dice is:

(a) 0.33 (b) 1
(c) 0 (d) 0.5

Answer: (c) 0 — a dice has only 1 to 6, so there is no number greater than 6

(ix) _____ outcomes are possible when we draw a card from deck of cards.

(a) 13 (b) 1
(c) 52 (d) 26

Answer: (c) 52 — a standard pack has 52 cards

(x) If P(E) = 0.07, then the probability of 'not E' is:

(a) 0.95 (b) 0.89
(c) 0.93 (d) 0.90

Answer: (c) 0.93 — P(not E) = 1 − 0.07 = 0.93

2. Arshia selects a day from (Tuesday, Wednesday) and a time from (Morning, Evening). Draw a possibility diagram. Also sketch a tree diagram.

Possibility diagram:

Day **Time**

	Morning	Evening
Tuesday	(Tue, Morning)	(Tue, Evening)
Wednesday	(Wed, Morning)	(Wed, Evening)

Possibility diagram of day and time

Tree diagram:



Tree diagram of day and time

Answer: S = {(Tue, M), (Tue, E), (Wed, M), (Wed, E)}, n(S) = 4

3. A card is drawn from a deck of cards. Find the probability of getting an ace or a spade card.

n(S) = 52

Aces: 4, so P(A) = 4 / 52

Spades: 13, so P(S) = 13 / 52

One card is both an ace and a spade, so P(A ∩ S) = 1 / 52

P(A ∪ S) = P(A) + P(S) − P(A ∩ S)

P(A ∪ S) = 4 / 52 + 13 / 52 − 1 / 52 = 16 / 52 = 4 / 13

Answer: P(ace or spade) = 4/13

4. Fatima dealt two cards successively (without replacement) from a shuffled deck of 52 playing cards. Find the probability that both cards are red.

There are 26 red cards in the pack.

P(first red) = 26 / 52 = 1 / 2

After one red card, 25 red cards are left out of 51 cards.

P(second red) = 25 / 51

P(both red) = 1 / 2 × 25 / 51 = 25 / 102

Answer: P(both cards red) = 25/102

5. If two dice are rolled, then find the probability of getting the product of face values 6 or the difference of face values 5.

n(S) = 36

A = product 6 = {(1, 6), (2, 3), (3, 2), (6, 1)}, n(A) = 4

B = difference 5 = {(1, 6), (6, 1)}, n(B) = 2

A ∩ B = {(1, 6), (6, 1)}, n(A ∩ B) = 2

P(A ∪ B) = 4 / 36 + 2 / 36 − 2 / 36 = 4 / 36 = 1 / 9

Answer: P(product 6 or difference 5) = 1/9

6. A bag contains 7 orange and 5 purple marbles. Two marbles are drawn one after the other without replacement. Find the probability that the first marble is orange and the second marble is also orange.

Total marbles = 7 + 5 = 12

P(first orange) = 7 / 12

After taking out one orange marble, 6 orange marbles are left out of 11.

P(second orange) = 6 / 11

P(both orange) = 7 / 12 × 6 / 11 = 42 / 132 = 7 / 22

Answer: P(both marbles orange) = 7/22

Quick Revision — Unit 12 at a Glance

P(E) = n(E) / n(S), 0 ≤ P(E) ≤ 1 and P(not E) = 1 − P(E).

Addition law: P(A ∪ B) = P(A) + P(B) − P(A ∩ B).

Multiplication law: without replacement, reduce the total.